

AI-based Defect Detection System that is Both High Performance and Highly Accurate Implemented in Low-cost, Low-power FPGAs

This paper presents a comparison between FPGA-based AI solution and GPU-based AI solution for industrial defect detection, showcasing FPGA's 24x higher accuracy, 80x better FPS/€, 20x lower power consumption, and 488x lower latency compared to GPU.

Authors

Leo Wiegand
One Ware, Co-Founder

Leon Beier
One Ware, Co-Founder

Walter Abramsohn
Altera, Device Portfolio Product
Marketing Manager

Rama Venkata
Altera, AI Senior Product Manager

Executive Summary

Altera's MAX[®] 10 FPGAs are widely deployed in industrial automation and robotics. Their low power, real-time processing, and cost efficiency make them an ideal platform for AI-powered defect detection.

FPGAs provide the flexibility and programmability needed to tailor designs and create custom AI networks from the ground up, delivering high performance and low latency. ONE WARE offers the ONE AI platform, which allows users to efficiently build and deploy customized AI solutions. Using ONE AI platform solutions and FPGAs flexibility, an AI-based defect detection system can be added in a tiny portion of a MAX 10 device without making extensive modifications to the existing manufacturing systems.

Industrial customers seeking to incorporate AI can build upon their existing FPGA designs and simply integrate AI functionality, without the need for new silicon architectures like GPUs. By adding AI directly to the FPGA, they can process data in real-time, ensuring high performance, low latency, and significant benefits in total cost of ownership (TCO) and time-to-market. Furthermore, power and cost-optimized FPGAs have lower power, smaller form factors, and longer lifecycles than GPUs.

Even a low-power, low-cost MAX 10 device featuring only 16K LEs could, when using ONE WARE's ONE AI quantization-aware training solution and INT8 optimization, achieve 99.5% accuracy and 1736 FPS at 0.5W, significantly outperforming GPU solutions like the Nvidia Jetson Orin Nano.

Introduction

The manufacturing industry is undergoing a major shift with Industry 5.0 with transformational technologies such as artificial intelligence (AI), machine learning, and robotics. One of the key aspects of manufacturing is defect detection so defective products do not reach customers and ensure manufacturers maintain quality and safety standards. The earlier you can catch defects in product manufacturing pipelines, the better, and manufacturers can keep costs down,

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preventing costly downtimes and product recalls.

While manual defect detection and visual inspection can be effective, automating the process using AI significantly improves accuracy and speed. AI-based systems analyze vast amounts of data, including images of defects and non-defective products. Over time, these systems can learn to distinguish between different types of defects, even those that are rare or subtle. This self-learning capability makes AI-powered solutions superior to traditional machine vision systems, especially in industries where defects vary widely.

Defect detection systems typically consist of high-resolution cameras, sensors, and lighting systems. These sensors collect real-time data from machines and devices on the factory floor, enabling highly accurate and consistent flaw detection enabling early intervention to address issues and reduce waste. Integrating AI into existing defect detection systems involves several challenges when using GPU-based solutions

- High power consumption of GPUs
- Latency issues with cloud-based solutions
- Costly infrastructure upgrade

To address these challenges, ONE WARE's ONE AI platform acts as an automatic AI developer, seamlessly generating tailor-made AI models with quantization-aware training (QAT) for defect detection, quality control, and root cause analysis that can be efficiently implemented in existing low-cost, low-power FPGAs such as Altera's MAX 10 or Cyclone® FPGA Families. ONE AI's quantization-aware training helps achieve higher accuracy and higher performance at reduced model sizes. This paper illustrates the development of an AI-based defect detection system that can go into an extrusion machine's quality control system using ONE WARE's ONE AI platform using an Altera MAX 10 FPGA.

The Challenge – Adding AI Capabilities Cost-effectively, Achieving Low Latency, High Performance, and Accuracy

The extrusion machine manufacturer in this case study already implemented a wide array of sensors, cameras, and radars and has Altera MAX 10 FPGAs on its machines. They wanted an efficient way to process and analyze the data generated by the sensors to detect defects using AI.

The challenge the manufacturer faced was to integrate AI capabilities without making extensive modifications to their existing systems. Their initial solution was to utilize a GPU-based AI solution, which is Nvidia Jetson Orin Nano, with a total power dissipation (TDP) of 10W. Upon evaluating the implementation of the AI model, they realized each machine was equipped with hundreds of sensors, making the cost of acquiring the necessary Jetson Nanos prohibitively expensive. Furthermore, there were space constraints, as each GPU required a cabinet on the factory floor, leading to further logistical and cost-related challenges. Sending the data from each sensor to GPU and getting back the feedback results in higher cost, higher power and higher latency solution.

Alternatively, transferring all the data from cameras and sensors to a central server location for processing and identifying defects is prohibitively expensive as the data is huge and would result in significant network infrastructure investment.

Solution – Implementing AI on MAX 10 FPGAs with ONE AI Platform

The existing hardware already contained Altera MAX 10 FPGAs with some unused logic resources. With ONE AI Platform a highly accurate and resource-efficient AI model could be built and deployed in the available unused logic elements of the MAX 10 FPGAs, seamlessly integrating AI capabilities directly into the machine's existing infrastructure. This allows the FPGAs to continue performing their primary function, while simultaneously taking on the AI data processing tasks.

ONE AI Platform solution helps users overcome the limitations of the Jetson Nano solution and helps integrate AI into existing systems without the need for costly hardware upgrades or compromising on sensor coverage. The result is a more efficient and cost-effective implementation of AI, enhancing the machine's capabilities without increasing the cost and space requirements of adding additional devices. ONE AI solution not only addresses user's immediate challenges but also provides a scalable path for future enhancements to the machines. By leveraging FPGA-based AI inference, designers can achieve real-time defect detection with lower latency, and improved reliability, ensuring superior performance in industrial automation applications.

ONE Minimalistic Quantization-Aware AI Model Implementation

To illustrate the design flow and advantages, ONE WARE chose a potato chip quality inspection example as a representative deep learning model for defect detection in extrusion equipment manufacturing use case they are servicing. The model detects burn marks on the potato chips. An opensource VGG19 image processing model pre-trained with dataset for doing potato chip quality inspection is available [on Github](#). ONE WARE used the same dataset on their minimalistic quantization-aware AI model generated by ONE AI tool targeting a MAX 10 device and compared the performance results.

The architecture designed by ONE AI is shown in Figure 1.

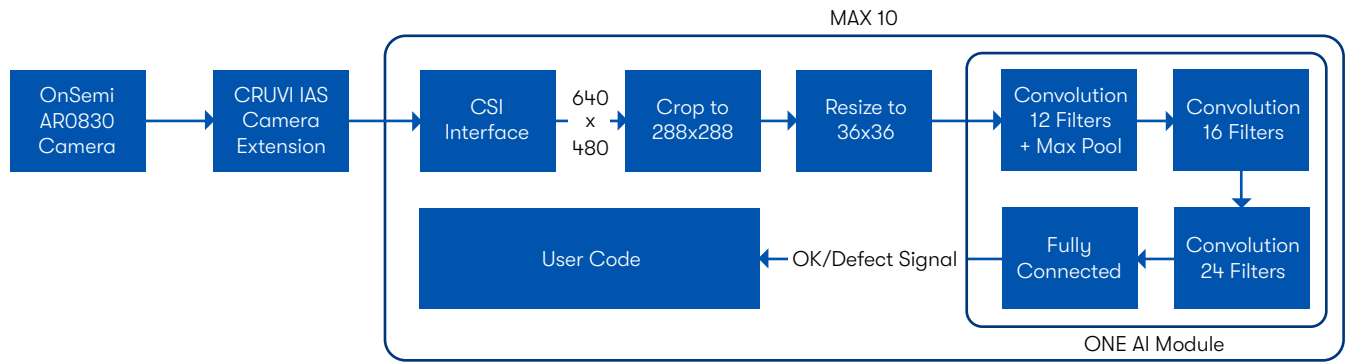


Figure 1. ONE AI Potato Chip Quality Inspection Deep Learning AI model

The AI model created by ONE AI is optimized by first cropping the 640x480 pixel input image to the central 80%x80% region and then resizing it to 36x36 pixels. This pre-processing step focuses the model on the most relevant area of the image while significantly reducing computational complexity.



Figure 2. Image of Defective chip



Figure 3. Image of good chip

The ONE AI model is trained and tested with INT8 quantization and simulated to determine accuracy. Quantization-Aware Training (QAT) is a crucial technique employed by ONE WARE's ONE AI solution to further enhance the efficiency and accuracy of AI models implemented on low-power Altera MAX 10 FPGAs. By simulating the quantization process during the model training phase, QAT helps ensure minimal loss of accuracy when converting from floating-point to fixed-point or INT8 formats, which is essential for hardware implementations. This approach allows the optimized models to significantly reduce computational requirements and memory footprint, enabling deployment on resource-constrained devices while maintaining high accuracy. QAT thus complements ONE AI's minimalistic architecture, making it an ideal method for achieving both high performance and low power consumption in industrial defect detection applications.

With ONE AI flow you can quickly test the accuracy of the model and latency of image processing in simulation itself without running the design on actual hardware. Users can set the performance target, and the tool optimizes the design for target FPS while improving the resource efficiency. The simulation time is extremely low so users can quickly iterate with different datasets and performance targets while keeping the resource utilization economical and efficient.

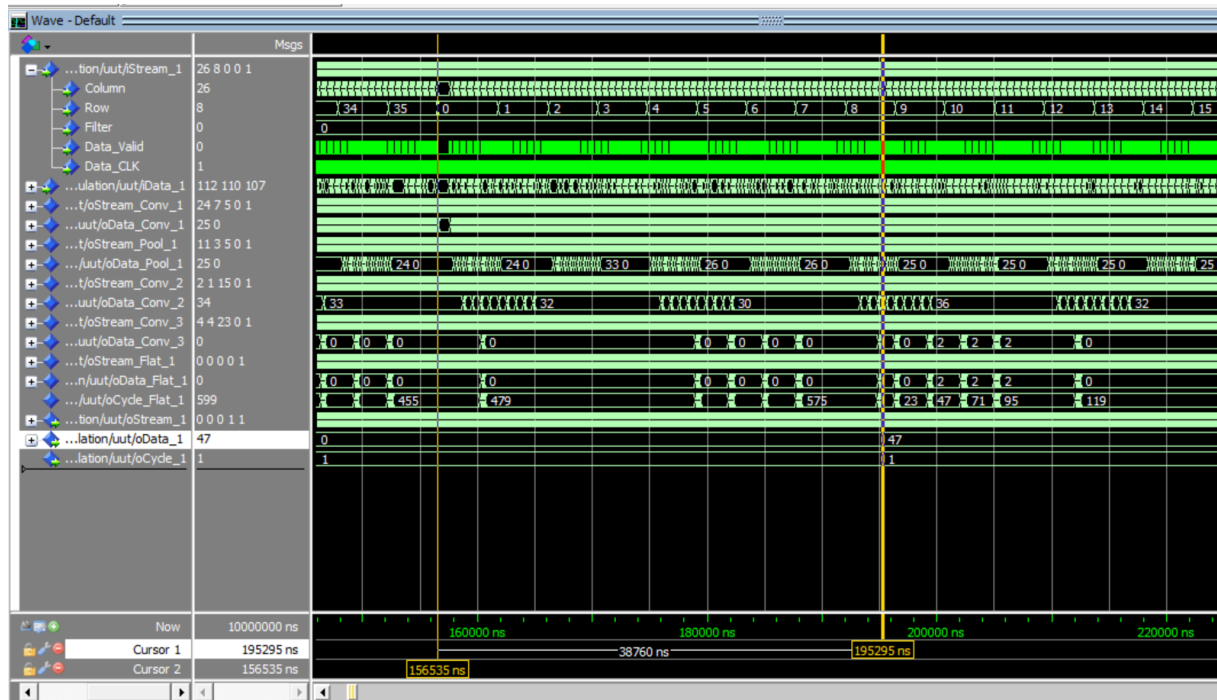


Figure 4. ONE AI model simulation

Table 1 shows the metrics of the VGG19 model implemented on Jetson Nano and ONE AI model implemented on Altera 10M16 FPGA. Validation Accuracy is derived from the running labelled validation dataset during training process to make sure the model is not overfitting. Test accuracy is derived from running unlabeled test dataset through the model once training is completed. This process is like deploying the model in the field.

Model	Validation Accuracy (%)	Test Accuracy (%)	GOPs	Parameters
VGG19	98 (FP32)	88 (FP32)	25	127 million
ONE AI	99.4 (INT8)	99.5 (INT8)	0.0175	6,750

Table 1. AI model metrics comparison between VGG19 model & ONE AI model

ONE AI's minimalistic architecture achieves higher efficiency by using only 0.0175 GOPs and 6,750 parameters, focusing exclusively on relevant features required for classifying good and defective potato chips. During testing ONE AI achieved 99.4% accuracy on validation images and 99.5% accuracy on test images with INT8 quantization.

In contrast, standard VGG19 model for this dataset, despite requiring significantly more computational resources (25 GOPs and 127 million parameters), achieved lower test accuracy (88% with FP32). The reason for this difference is overfitting in the larger model. It memorizes a large amount of irrelevant information from the training data, which is unnecessary for classifying defective potato chips. This results in less reliable performance on new, unseen data.

The ONE AI model and image pre-processing logic are exported to RTL. Table 2 shows sample resource utilization of the AI model on 10M16 for different performance targets.

Sample ONE AI performance targets	Logic Elements Used	Memory Used	DSP Used	Max. Fmax	Max. Frames per Second (FPS)
Efficient	2.7 K (17%)	8.9 KB (13%)	4 (4%)	39 MHz	70 FPS
Balanced	3.5K (22%)	8.9KB (13%)	17(19%)	45 MHz	479 FPS
FAST	5.5 KLE (39%)	8.9KB (13%)	59 (66%)	45 MHz	1736 FPS

Table 2. Resource utilization for each ONE AI performance targets

The RTL design is compiled in Quartus® Prime software targeting 10M16 design.

Entity:Instance	Logic Cells	ed Logic R	I/O Registers	Memory Bits	M9Ks	I Bl	DSP Elements	DSP 9x9	I
MAX 10: 10M16SAU169C8G									
AI_Demo	5520 (4)	2931 (1)	0 (0)	70968	30	1	59	59	0
CNN:CNN1	5516 (33)	2930 (19)	0 (0)	70968	30	0	59	59	0
CNN_Convolution:CNN_Convolution_1	1521 (965)	754 (542)	0 (0)	6048	6	0	18	18	0
CNN_Convolution:CNN_Convolution_2	2068 (11...	1079 (558)	0 (0)	20736	10	0	24	24	0
CNN_Convolution:CNN_Convolution_3	1606 (595)	825 (194)	0 (0)	32856	11	0	16	16	0
CNN_Pooling_Efficient:CNN_Pooling_1	108 (108)	103 (103)	0 (0)	1728	1	0	0	0	0
NN_Layer:NN_Layer1	191 (191)	150 (150)	0 (0)	9600	2	0	1	1	0

Figure 5. Quartus Hierarchy Window showing the ONE AI model resource utilization for Fast target

Figure 6, Figure 7 and Figure 8 show the Quartus® Prime software RTL view of ONE AI model implementation in a MAX 10 FPGA.

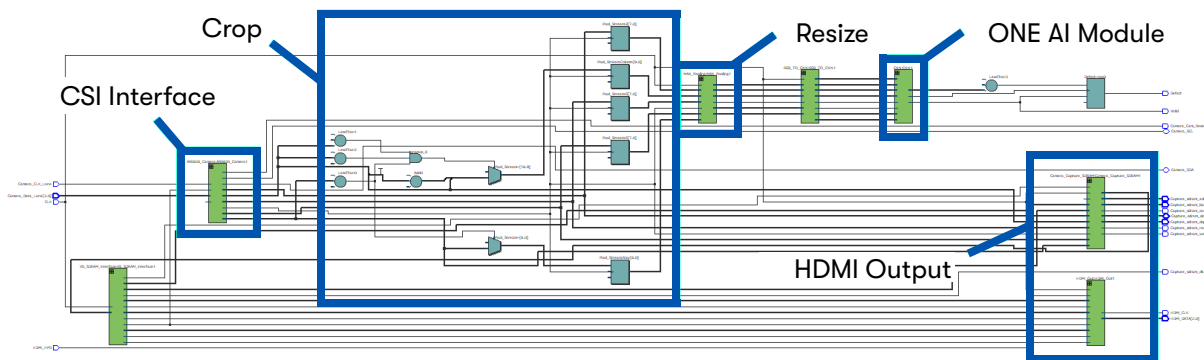


Figure 6. Top level view with camera interface, scaling of image from 640x480 to 36x36 pixels, AI, SDRAM buffer and HDMI output

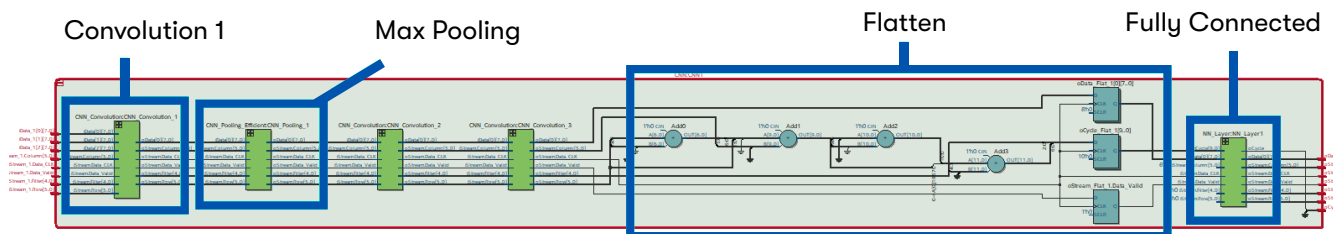


Figure 7. AI with 3 convolutions, 1 pooling and 1 dense layer

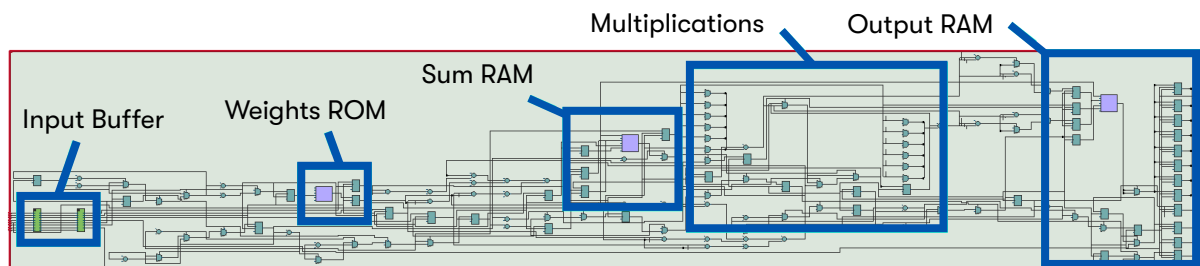


Figure 8. Parallel implementation of convolution

Once the design goes through the push-button Quartus compilation, a bitstream is generated and programmed to the hardware. Users can connect the camera input stream and feed the programmed model to detect the defective/ok output.

Results – MAX 10 FPGAs Outperform Jetson Orin Nano in Power, Cost, Latency, and Performance

Here is a comparison of the accuracy, power, cost, latency, and performance of ONE AI model implementation on a 10M16 density device and Jetson Orin Nano.

	Altera MAX 10 FPGA	Nvidia Jetson Orin Nano ¹	FPGA Benefit vs GPU
AI Model	ONE WARE's ONE AI model	VGG19	
Test Accuracy	99.5% (INT8)	88% (FP32) ²	24x higher accuracy ⁹
Power	0.5W	10W ³	20x lower power
Size	11x11 mm	70x45 mm	26x smaller footprint
Cost	€45 ⁴	€250 ⁵	6x lower cost
Performance	1736 FPS (INT8)	24 FPS (FP16) ⁶	72x higher FPS
Latency	0.086 ms ⁷	42 ms ⁸	488x lower latency

Table 3. Comparison of AI Benchmark Results

In this example, MAX 10 is 6x less expensive than Jetson Nano, providing 488x lower latency processing speeds at 20x lower power achieving 24x higher accuracy.

FPGAs with DSP blocks for compute offer customizable precisions, and customizable memory structures offer storing of data on-chip or off-chip, which are all key to implementing optimal AI networks. Support for multiple I/O standards enable direct connection to diverse sensors. Overall, FPGAs offer flexibility and programmability to customize your design and build any AI network from ground up to get best performance and latency.

GPUs are general-purpose devices and can run a variety of AI workloads. But the efficiencies are not optimal as they are optimized for popular AI models like Gen AI LLMs.

MAX 10 FPGAs Ideal for Industrial Applications

MAX 10 devices are unique in the market due to their high integration of features that are specifically targeting industrial machinery. The devices also use PLLs, as opposed to the common DLLs and allow a higher tolerance in jitter which is crucial in noisy industrial environments. MAX 10 devices with single / dual voltage supply are voltage sequence agnostic making any card using them hot swappable which is great advantage in industrial use cases.

They have hardened multipliers embedded in the device making them suitable for AI applications. MAX 10 is the only flash-based FPGA device with embedded DSP blocks in power and cost-optimized range of devices.

The MAX 10 has up to two analog-to-digital converters, and each has up to 17 channels that they can measure. This is ideal for sensors in the industrial world. The lifecycle for MAX 10 devices has been extended up to 2040, ensuring long life and reliable supply.

The devices offer up to 500 I/Os in high-density packages. The smallest package is 3x3mm. Here is the summary of [Altera MAX 10 device features](#).

¹ The Jetson Orin Nano module requires a custom mainboard, just like the Altera MAX[®] 10 FPGA Chip

² On the GitHub repository, FP16 training was not enabled for TensorFlow, so the model should be trained with FP32

³ TDP: <https://www.techpowerup.com/gpu-specs/jetson-nano.c3643>

⁴ Excluding VAT: <https://www.digikey.de/de/products/detail/altera/10M16SAU169C8G/5284834>

⁵ Excluding VAT: <https://www.digikey.de/de/products/detail/seeed-technology-co-ltd/102110800/20372495>

⁶ FP16 was chosen because it is a common option for inference. VGG19 (224×224) on Jetson Nano achieves 8.9 FPS (<https://openbenchmarking.org/result/19Q8087-HV-19Q8018HV82>). The Jetson Nano has 0.47 TFLOPs FP16 (<https://www.techpowerup.com/gpu-specs/jetson-nano.c3643>), whereas the Jetson Orin Nano has 1.28 TFLOPs FP16 (<https://www.techpowerup.com/gpu-specs/jetson-orin-nano-4-gb.c4083>). Based on this, VGG19 (255×255) should run at about 24 FPS on the Jetson Orin Nano.

⁷ Latency is calculated as the time between reception of the last pixel from the camera and computation of the AI result. Since the FPGA processes the AI simultaneously while receiving pixels, its latency is lower than that of the Jetson Orin Nano.

⁸ Typically, an image is first stored in RAM before AI processing begins. Thus, the minimum latency should be approximately 1 / (24 FPS).

⁹ Orin nano - 12 defects/100 samples; ONE AI - 0.5 defects/100 samples. Accuracy = 12/0.5 = 24x higher accuracy.

Conclusion

With ONE WARE's ONE AI Platform solution, AI can be integrated into existing systems with FPGAs without costly hardware upgrades or compromising on sensor coverage.

ONE AI tailor-made AI solution with quantization-aware training not only achieves higher accuracy but also enables efficient deployment on low-performance hardware, making it an ideal solution for real-time quality control, condition monitoring, and computer vision tasks in industrial applications.

ONE AI's optimized FPGA-based AI delivers superior defect detection without costly hardware upgrades. By integrating AI directly into existing FPGA infrastructure, manufacturers can achieve higher accuracy, lower costs, and real-time performance—making it an ideal fit for adding AI into industrial automation applications.

Learn More:

- [1] [One Ware](#)
- [2] [Altera Product Families](#)
- [3] [Recommended MAX 10 board](#)
- [4] [Other MAX 10 boards](#)
- [5] [One Ware, a new Altera Solution Acceleration Partner](#)



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